

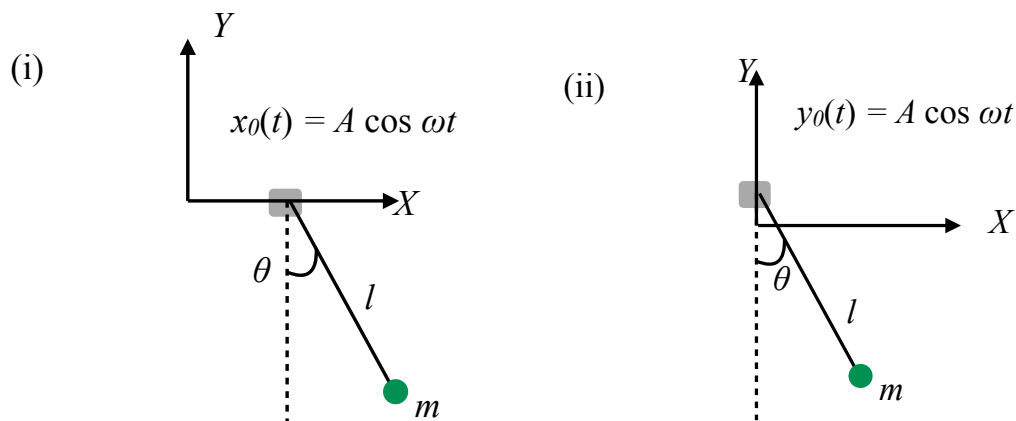
Classical Mechanics - Home Assignment 1

Lagrange's Equation (August 2024)

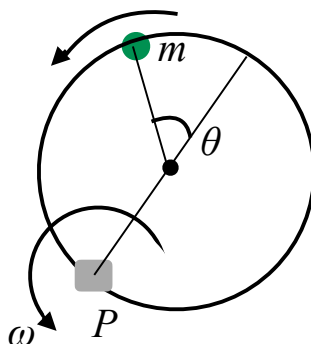
1. Consider a charged spinless particle moving in an electromagnetic field $(\vec{E}, \vec{B})$. The force acting on the particle is given by $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. The scalar and vector potentials related to the field are given by $\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$, and $\vec{B} = \nabla \times \vec{A}$. Show that Lagrange's equations are satisfied for a potential function

$$U = q\left(\phi - \vec{v} \cdot \vec{A}\right)$$

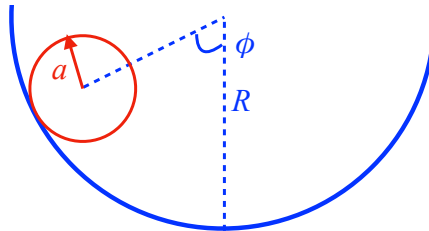
2. A pendulum consists of a mass m and a massless stick of length ℓ , hanging from a support. Derive Lagrange's equation of motion of the mass, when the support oscillates with amplitude A and angular frequency ω in the (i) horizontal, and (ii) vertical direction.



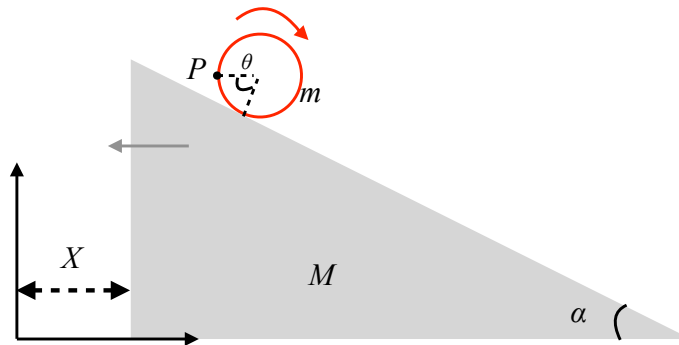
3. A bead of mass m slides on a smooth circular loop of radius a placed on a horizontal plane. The loop itself rotates in the horizontal plane about a point P on the circumference with constant angular velocity ω . Find the motion of the bead using Lagrange's equation of motion.



4. A uniform cylinder of radius a and mass m is rolling (without slipping) under gravity inside a bigger (fixed) cylinder of radius R . Find the motion of the moving cylinder.



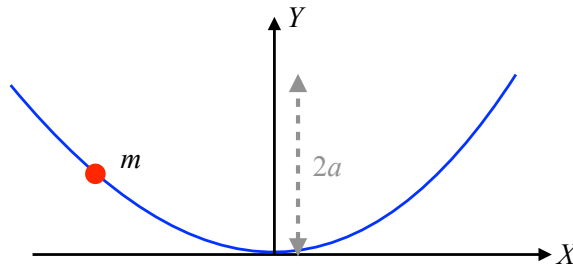
5. A uniform sphere of radius a and mass m is rolling (without slipping) under gravity down a triangular block of mass M , that is free to move on a frictionless horizontal surface. Determine the motion of the system, given that initially all objects are at rest and the centre of the sphere is at a distance h above the horizontal surface.



6. A bead of mass m slides on a frictionless wire in the shape of a cycloid

$$x = a(\theta - \sin \theta) \quad y = a(1 + \cos \theta)$$

with $-\pi \leq \theta \leq \pi$. Derive the equation of motion and find the solution.



Solutions

Problem 1

We have the Lagrangian

$$L = \frac{1}{2}m\dot{x}_i\dot{x}_i - q(\phi - \dot{x}_i A_i) \quad (1)$$

Now use Lagrange Equation of motion

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_j} \right) - \frac{\partial L}{\partial x_j} &= 0 \\ \frac{d}{dt}(m\dot{x}_j + qA_j) + q \left(\frac{\partial \phi}{\partial x_j} - \dot{x}_i \frac{\partial A_j}{\partial x_i} \right) &= 0 \\ F_j + q(\partial_t A_j + \dot{x}_i \partial_i A_j) + q(\partial_j \phi - \dot{x}_i \partial_i A_j) &= 0 \end{aligned}$$

$$\begin{aligned} F_j &= -q(\partial_t A_j + \dot{x}_i \partial_i A_j) - q\dot{x}_i (\delta_{ki} \delta_{lj} - \delta_{kj} \delta_{li}) \partial_k A_l \\ &= -q(\partial_t A_j + \dot{x}_i \partial_i A_j) - q\dot{x}_i \epsilon_{pkl} \epsilon_{pij} \partial_k A_l \\ &= -q(\partial_t A_j + \dot{x}_i \partial_i A_j) + q\epsilon_{jip} \dot{x}_i (\nabla \times A)_p \end{aligned}$$

now use $E_j = -\partial_j \phi - \partial_t A_j$ and $B_j = \epsilon_{jlm} \partial_l A_m$ And hence

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (2)$$

Problem 2

Solution:

(i)

We have $x = A \cos \omega t + l \sin \theta$ and $y = -l \cos \theta$ therefore

$$\begin{aligned} \dot{x} &= -A\omega \sin \omega t + l \cos \theta \dot{\theta} \\ y &= l \dot{\theta} \sin \theta \end{aligned}$$

Hence we can write the Lagrangian as

$$\begin{aligned} L &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy \\ &= \frac{1}{2}m \left(A^2 \omega^2 \sin^2 \omega t + l^2 \dot{\theta}^2 - 2Al\omega \cos \theta \sin \omega t \dot{\theta} \right) + mgl \cos \theta \end{aligned}$$

(ii)

Here $x = l \sin \theta$ and $y = A \cos \omega t - l \cos \theta$

Therefore,

$$\dot{x} = l \cos \theta \dot{\theta}$$

$$\dot{y} = l \sin \theta \dot{\theta} - A \omega \sin \omega t$$

$$L = \frac{1}{2} m \left(l^2 \dot{\theta}^2 + A^2 \omega^2 \sin^2 \omega t - 2 A l \omega \sin \theta \dot{\theta} \sin \omega t \right) - m g (A \cos \omega t - l \cos \theta)$$